

An Introduction to Data Forensics & Using it to Improve Test Security



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Disclosure of Conflicts of Interest

We have nothing to disclose





Agenda

- Data Forensics overview
- How ABS uses Data Forensics
- How ABIM uses Data Forensics



Data Forensics Overview

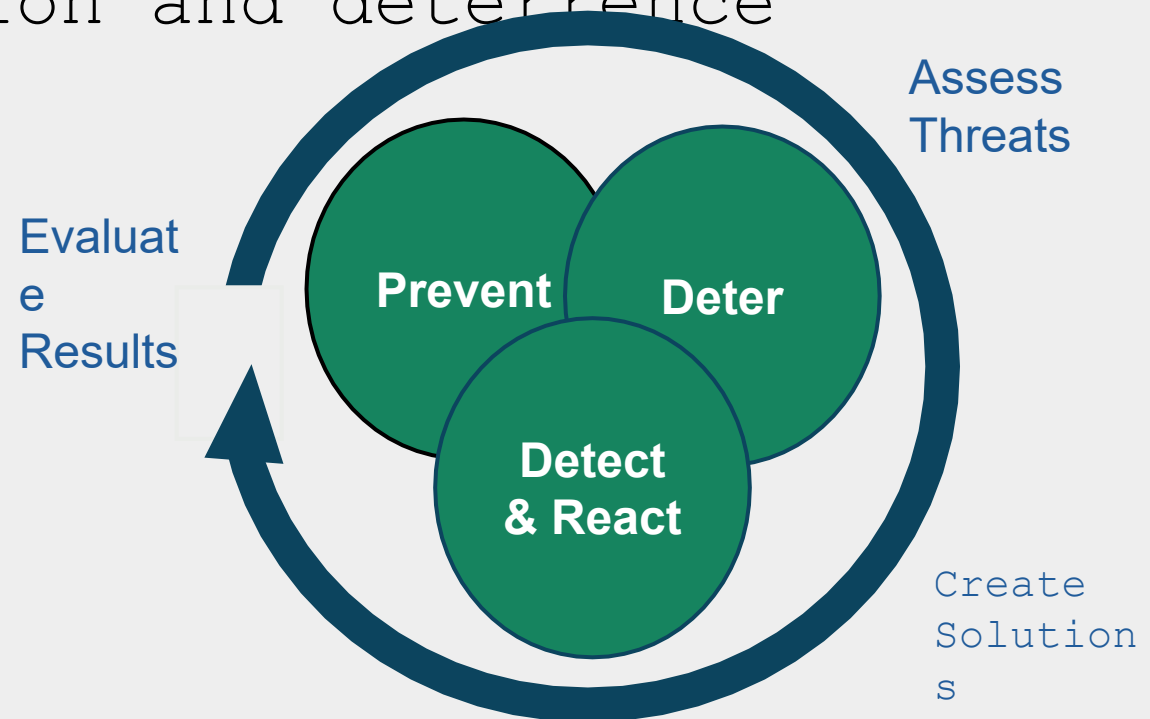


Definition of Data Forensics

- Data Forensics is the statistical analysis of test result data to determine whether test security violations potentially occurred
 - Functions as a quality assurance measure to determine if test scores are valid and trustworthy
 - Functions as a security measure to detect anomalies that may prompt action
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Data Forensics in the Test Security Process

- Firmly in “Detect & React”
- Data Forensics findings can inform decisions regarding prevention and deterrence





What Does Data Forensics Detect?

- Data Forensics detects anomalies in the test data that are associated with potential test security issues
- Data Forensics does not detect cheating
 - Cheating is a behavior and requires intent
 - As measurement professionals, our focus should be on score validity
 - Do the anomalies cause us to question score validity?



Scientific Defensibility

- Analyses are based on statistical theory and accepted scientific principles
 - Results are credible
 - Results are factual
- Probabilities are measured objectively
- Conservative thresholds give strong, compelling results



Legal Defensibility

- Possible actions based on Data Forensics results should be outlined in the candidate agreement
- Data Forensics results are circumstantial evidence
 - Requires an inference or deduction
 - Plausible disconfirming explanations should be considered
- Score invalidations based on Data Forensics have been upheld in court



Common Data Forensics Analyses

- Response similarity
- Policy violations
- Response changes
- Score differencing
- Response times
- Score gains
- Person fit/aberrance



Response Similarity

- Tests are compared pairwise to determine the probability of the counts of matching correct & incorrect responses
 - Clusters are formed from the pairwise results
 - Are the tests identical?
 - Extreme similarity/identical tests can result from collusion, using the same braindump, answer copying, proxy test takers, etc.
 - Sufficiently strong to support score invalidation
- 



Policy Violations

- Did the examinee violate the retake policy in their attempts?
- Score invalidation is justified if the rules were broken



Answer Changes

- Did the examinee have a high rate of wrong-to-right (WTR) answer changes?
- Did the examinee have significantly more WTR answer changes than right-to-wrong (RTW)?
- Can be especially telling if the examinee took an unscheduled break before the answer changes
- Can be evidence of tampering or reviewing material during the testing session



Score Differencing

- Examinees should have comparable performance between item subsets
- Significantly better performance on one subset may be evidence of preknowledge of those items
- Examples:
 - Scored items vs non-scored/field test items
 - Old items vs new items
 - High-exposure items vs low-exposure items
 - Braindump items vs non-braindump items



Response Times

- Does the examinee have rapid response times?
- Do the examinee's response times correlate with those of other examinees?
- Can be evidence of test security threats
 - Anomalous response times with high scores may be evidence of preknowledge or proxy test takers
 - Anomalous response times with low scores may be evidence of item harvesting



Score Gains

- Does the examinee have large changes in scores between attempts?
- Length of time between attempts is an important factor
- Can be evidence of braindump use or hiring a proxy test taker



Person Fit/Aberrance

- Detects tests that do not fit the statistical models
 - Correctly answer difficult items but miss easier items
 - Test score is improbable based on ability estimate
 - Examinee is better modeled as having two ability levels rather than one
- Not the strongest evidence of test security threats
- Useful for analyzing items



Benefits of Data Forensics

- You can't manage what you don't measure
- Data Forensics results can
 - Identify invalid scores
 - Identify items to be retired or refurbished
 - Inform your republication schedule
 - Inform test security investigations
 - Evaluate the efficacy of test security measures



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How ABS Uses Data Forensics

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Data Forensics & ABS Examinations

- ABS In-Training Examination (ABSITE)
 - Low- to medium-stakes
 - Intended to provide feedback to programs on trainee strengths/weaknesses
 - Often used for fellowship applications, etc.
 - Proctored delivery at residency programs, via ITS, 4-5 day admin window
- General Surgery Qualifying Examination
 - High stakes: First step in certification process
 - Secure delivery via Pearson VUE centers, typically on a single day
- Other exams on as-needed basis (typically due to extended admin windows)
 - 2026 Specialty Qualifying Exams
 - FPDs

What We Provide

- Examinee information
 - ID, test site, test date, training level
- Item-level information
 - Examinee response, correct response, number of visits & response time
 - New v. used item
- What we *don't* provide:
 - Examinee names
 - Seating chart

Response Similarity – Course of Action

- Review analysis results
- Examine circumstances of flagged clusters
 - Test date & site(s)
 - Seating chart
 - Avg response time & number of visits
 - Performance
- Notify program(s) of flagged examinees & level of similarity

Response Similarity – Course of Action

- Program(s) required to investigate the situation & report back to ABS
 - Interviews with residents
 - Room layout & proctoring
 - Suspicious behavior
 - Adherence to policies re: cell phones, bathroom breaks, laptops/internet
- Withhold scores
 - We hold results of all trainees at program until investigative report received
 - ABS does not consider results for flagged examinees to be valid and will not release them
- *Consequences to individuals left up to program*

Other Findings

- Response time
 - Typically, low effort
 - Review past/future performance (harvesting?)
- New-Old Difference
 - Individual AND test site
 - 2023 exam
 - 23 individual test instances flagged
 - One test site had 14 of 37 instances flagged
 - Grand Rounds ABSITE review given by external source
 - Content review suggested the review course contained exam content

Implications of Data Forensics Results

- Reinforces decision to randomize item order
 - ABSITE not randomized one year
 - 10 flagged clusters (compared to ~3-5)
- Beginning with 2027 ABSITE, completely new items
 - Evidence of content sharing/item banks
 - New-Old analysis from 2022 ABSITE
 - 2025 ABSITE – program investigation into similar yielded a shared document circulated among residents including items from prior years

Future Directions

- Continuing use of data forensics
- Expanded set of analyses
 - Item response history now available for ABSITE
- Possible implications on specialty exam administration date(s)



How ABIM Uses Data Forensics

Data Forensics and ABIM Assessments

- ABIM long form assessments (Initial Certification and Maintenance of Certification)
 - Proctored delivery at test center
 - Primary security threat is item preknowledge
- ABIM Longitudinal Knowledge Assessments (LKA)
 - Non-proctored delivery on diplomates' personal devices
 - Code of conduct prohibits getting help from any other person, reproducing or disclosing exam content, and copying exam content verbatim into any AI tool
 - Primary security threats are item preknowledge, collusion, and unauthorized use of AI tools

Taking Action on Statistical Flags

- Hold scores for candidates who meet predefined flagging criteria
- Notify candidates via letter
 - Request information about how they prepared for the exam
 - Conduct content review of study materials
- Interview candidates
- Present findings to policy board to determine course of action, which may include score invalidations
- Score invalidations may be based on statistical evidence alone, but we do not accuse physicians of “cheating”

Case Study 1: Flagged pair

- Initial flag for response similarity
- Same residency program

Name	Test Date	Test Score	Test Score: Matching Items	Test Score: Nonmatching Items
Diplomate 1	Date 1	79%	82%	68%
Diplomate 2	Date 2	68%	82%	23%

Case Study 2: Large cluster

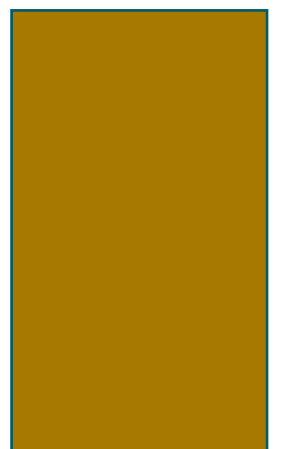
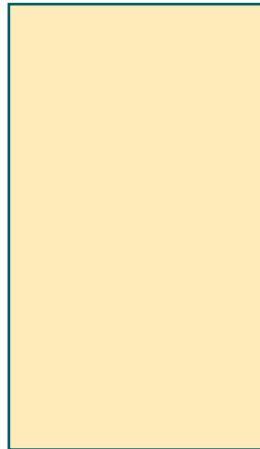
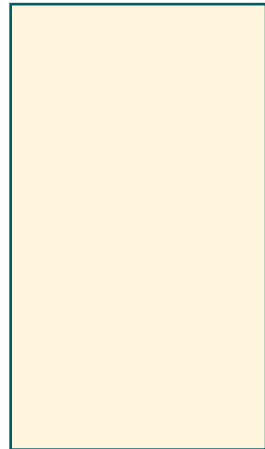
- Response similarity is conducted across all pairs of examinees, but pairs can be part of a larger cluster
- In 2023, a large cluster of examinees was flagged in the Internal Medicine Initial Certification exam
- ABIM invalidated results for 62 physicians

Case Study 2: Assessment Design

62 common anchor items
administered to every
physician



178 unique items that
physicians may or may
not have in common
based on the form they
were administered



Case Study 2: Performance and Response Times

Taker	Exam Date	Item Grouping	Average Time (s)	Percent Correct
A	2023-08-22	Common	80	0.97
		Unique	130	0.72
B	2023-08-22	Common	65	1.00
		Unique	132	0.76
C	2023-08-28	Common	89	0.92
		Unique	125	0.69
D	2023-08-28	Common	63	0.92
		Unique	106	0.60
E	2023-08-28	Common	54	0.98
		Unique	69	0.78
F	2023-08-28	Common	99	0.95
		Unique	125	0.73

On average these physicians spent **40 seconds less** on common items and performed **25% higher** on common items compared to unique items.

Summary

- ABIM utilizes a variety of data forensic techniques to monitor for testing irregularities
- Data forensics is only a first step in a comprehensive plan to take action
 - Additional context
 - Content reviews
 - Interviews
 - Policy board

Questions?

Thank you!

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